

Routing – Chris Bryant Course

Classful Ranges	
A	0.0.0.0 – 127.255.255.255
B	128.0.0.0 – 191.255.255.255
C	192.0.0.0 – 223.255.255.255
D	224.0.0.0 – 239.255.255.255
E	240.0.0.0 – 255.255.255.255

Loopbacks (127.x.x.x)

127.0.0.0 is reserved for host loopbacks, not for cisco router loopback interfaces.

127.0.0.1 – on windows is localhost

Class D: First octet of 224-239. Reserved for multicasting

Multicast – a multicast is traffic destined for a particular group of hosts. It's a middle ground between unicasts and broadcasts.

Class E: First Octet 240-255. Reserved for Future Use... "experimental addresses."

RFC 1918 Reserved Private Address Space

Reserved Ranges	
RFC 1918	10.0.0.0 – 10.255.255.255
Localhost	127.0.0.0 – 127.255.255.255
RFC 1918	172.16.0.0 – 172.31.255.255
RFC 1918	192.168.0.0 – 192.168.255.255

Private address space is reserved for internal networks. Not globally routable. Private address must be NATed to become a routable. NAT and/or PAT (Port Address Translation)

10.0.0.0 /8

172.16.0.0/12

192.168.0.0/16

Routing

If the destination is on the same subnet as the source, the router really doesn't need to get involved. Here, Host A is sending packets to 192.168.1.15 /24, a host on the same subnet (192.168.1.0/24). No routing is necessary. In effect, these packets go straight to 192.168.1.15.

